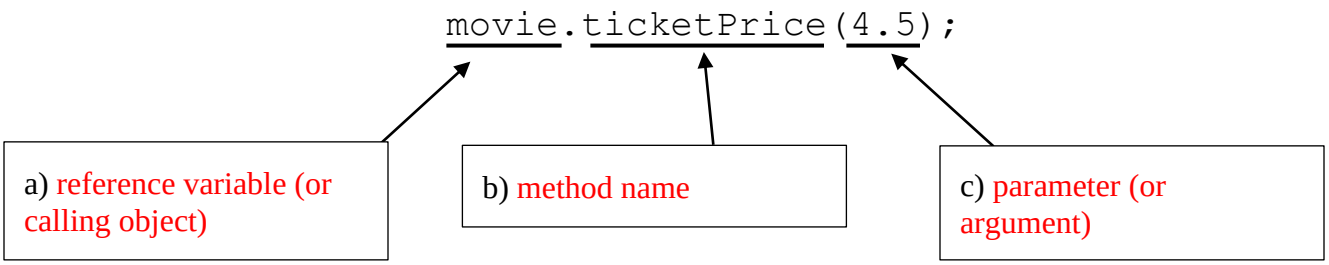
Unit 1 – Using Objects and Methods Name: **KEY**
HW 6

1) Label the following pieces of the method call.



2) Consider the following overloaded methods.

```
public double example(int a, boolean b)
{
     /* implementation not shown */
}
public double example(int a, int b)
{
     /* implementation not shown */
}
public double example(boolean a, int b)
{
     /* implementation not shown */
}
```

Which of the following lines of code, if located in a method in the same class as `myMethod`, will compile without error? Circle your answer.

`double outcome = example(false, 3.5);`	Legal	Illegal
`int outcome = example(5, false);`	Legal	Illegal
`double outcome = example(3, 3);`	Legal	Illegal
`int outcome = example(5.5, true);`	Legal	Illegal
`double outcome = example(0, false);`	Legal	Illegal
`double outcome = example(true, 15);`	Legal	Illegal
`double outcome = example(5.5, true);`	Legal	Illegal

3) Consider the following class declaration

```
public class MyClass {
     private int num;

     public MyClass() {
          num = 5;
     }

     public MyClass(int n) {
          num = n;
     }

     public void increment(int more) {
          num = num + more;
     }

     public int getNumValue() {
          return num;
     }
}
```

The following segments of code appear in another class. Determine their output.

a)

```
MyClass one = new MyClass();
MyClass two = new MyClass(20);
one.increment(20);
System.out.println(one.getNumValue() + " " + two.getNumValue());
```

Output: 25 20

b)

```
MyClass three = new MyClass(5);
int four = three.getNumValue() + 10 % 3;
System.out.println(four);
```

Output: 6

c)

```
MyClass five = new MyClass();
MyClass six = new MyClass(30);
six.increment(10);
System.out.println(five.getNumValue() + " " + six.getNumValue());
```

Output: 5 40

4) Create a `Circle` class that contains the following

- One instance variable – radius
- A default constructor
- A parameter constructor
- A method named `initialize` that prompts the user for a radius value and then assigns the entered value to the radius instance variable.
- A method named `printCircleData` that used the circle's radius to calculate the circle's diameter, circumference, and area.

```
import java.util.Scanner;
public class Circle
{
  private int radius;

  public Circle(){
    radius = 1;
  }

  public Circle(int c){
    radius = c;
  }

  public void initilaize(){
    Scanner keyboard = new Scanner(System.in);
    System.out.println("Please enter the radius: ");
    radius = keyboard.nextInt();
  }

  public void printCircleData(){
    int diameter = 2 * radius;
    double cir = 2 * 3.14 * radius;
    double area = radius * radius * 3.14;
    System.out.println("Diameter: " + diameter);
    System.out.println("Circumference: " + cir);
    System.out.println("Area: " + area);
  }
}
```